第二章自测题

题量: 35 满分: 100 作答时间: 10-12 10:21 至 10-27 13:00

一. 单选题（共35题，100分）

1. (单选题) If the bandwidth of a signal is 5 KHz and the lowest frequency is 52 KHz, what is the highest frequency?

A 5 KHz

B 10 KHz

C 47 KHz

D 57 KHz

2. (单选题)
A signal is measured at two different points. The power is P1 at the first point and P2 at the second point. The dB is 0. This means _____.

A A. P2 is zero

B P2 equals P1

C P2 is much larger than P1

D P2 is much smaller than P1

3. (单选题) For a _____ channel, the Nyquist bit rate formula defines the theoretical maximum bit rate.

A noisy

B noiseless

C bandpass

D low-pass

4. (单选题) For a _____ channel, we need to use the Shannon capacity to find the maximum bit rate.

A noisy

B noiseless

C bandpass

D low-pass

5. (单选题) In _____ transmission, a start bit and a stop bit frame a character byte.

A asynchronous serial

一. 单选题（100分）
1 2 3
6 7 8
11 12 13
16 17 18
21 22 23
26 27 28
31 32 33

C parallel

D (a) and (b)

6. (单选题) In _____ transmission, we send bits one after another without start or stop bits or gaps. It is the responsibility of the receiver to group the bits.

A synchronous

B asynchronous

C isochronous

D none of the above

7. (单选题) Microwaves are _____.

A omnidirectional

B unidirectional

C bidirectional

D none of the above

8. (单选题) Radio waves are _____.

A omnidirectional

B unidirectional

C bidirectional

D none of the above

9. (单选题) Which of the following primarily uses guided media?

A cellular telephone system

B local telephone system

C satellite communications

D radio broadcasting

10. (单选题) _____ are used for short-range communications such as those between a PC and a peripheral device.

A Radio waves

B Microwaves

C Infrared waves

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11. (单选题) If the available channel is a ____ channel, we cannot send a digital signal directly to the channel.

A low-pass

B bandpass

C low rate

D high rate

12. (单选题) Baseband transmission of a digital signal is possible only if we have a ____ channel.

A low-pass

B bandpass

C low rate

D high rate

13. (单选题) In Manchester and differential Manchester encoding, the transition at the middle of the bit is used for _____.

A bit transfer

B baud transfer

C synchronization

D none of the above

14. (单选题) PCM is an example of _____ conversion.

A digital-to-digital

B digital-to-analog

C analog-to-analog

D analog-to-digital

15. (单选题) The minimum bandwidth of Manchester and differential Manchester is ____ that of NRZ.

A the same as

B twice

C thrice

D none of the above

16. (单选题) _____ conversion involves three techniques: line coding, block coding, and scrambling.

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B Digital-to-analog

C Analog-to-analog

D Digital-to-digital

17. (单选题) If the baud rate is 400 for a QPSK signal, the bit rate is _____ bps.

A 100

B 400

C 800

D 1600

18. (单选题) In QAM, both _____ of a carrier frequency are varied.

A frequency and amplitude

B phase and frequency

C amplitude and phase

D none of the above

19. (单选题) ASK, PSK, FSK, and QAM are examples of _____ conversion.

A digital-to-digital

B digital-to-analog

C analog-to-analog

D analog-to-digital

20. (单选题) We can divide ___ into two different schemes: synchronous or statistical.

A FDM

B TDM

C WDM

D none of the above

21. (单选题) Which multiplexing technique transmits analog signals?

A FDM

B TDM

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21 22 23

26 27 28

31 32 33

D (a) and (c)

22. (单选题) Which multiplexing technique transmits digital signals?

A FDM

B TDM

C WDM

D None of the above

23. (单选题) _____ can be applied when the bandwidth of a link (in hertz) is greater than the combined bandwidths of the signals to be transmitted.

A TDM

B FDM

C Both (a) or (b)

D Neither (a) or (b)

24. (单选题) _____ conversion is the process of changing one of the characteristics of an analog signal based on the information in the digital data.

A Digital-to-analog

B Analog-to-analog

C Analog-to-digital

D Digital-to-digital

25. (单选题) _____ is the set of techniques that allows the simultaneous transmission of multiple signals across a single data link.

A Demodulating

B Multiplexing

C Compressing

D None of the above

26. (单选题) A _____ network is a cross between a circuit-switched network and a datagram network. It has some characteristics of both.

A virtual-circuit

B packet-switched

C frame-switched

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27. (单选题) A local telephone network is an example of a _____ network.

A packet-switched

B circuit-switched

C message-switched

D none of the above

28. (单选题) A traditional cable TV network transmits signals _____.

A upstream

B downstream

C upstream and downstream

D none of the above

29. (单选题) Data from a computer are _____; the local loop handles _____ signals.

A analog; analog

B analog; digital

C digital; digital

D digital; analog

30. (单选题) In _____, the resources need to be reserved during the setup phase; the resources remain dedicated for the entire duration of data transfer phase until the teardown phase.

A datagram switching

B circuit switching

C frame switching

D none of the above

31. (单选题) Packet-switched networks can also be divided into _____subcategories: virtual-circuit networks and datagram networks

A four

B three

C two

D five

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A voice communication

B upstream data

C downstream data

D control data

33. (单选题) The network layer in the Internet is designed as a _____ network.

A virtual-circuit

B datagram

C circuit-switched

D none of the above

34. (单选题) The standard for data transmission over an HFC network is called _____.

A MCNS

B DOCSIS

C CMTS

D ADSL

35. (单选题) The term modem is a composite word that refers to the two functional entities that make up the device: a signal _____ and a signal _____.

A modulator; demodulator

B demodulator; modulator

C modern; demo

D none of the above

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